Computer Engineering Department

Spring 2024 **EC372 Midterm Exam** **Time: 90 min**

Student Name:…… ……... Student ID: …….

Use 4 significant digits in your calculations.

Q1) Use Taylor polynomial of the function $f(x) = \sqrt{x}$ from a = 2.0 to find *f*(2.5) for a relative true error of less than 0.05%. **(14 marks)**

$$f(a) + f'(a)(x-a) + \frac{f^{(2)}(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \ldots$$

Q1)

$F(x) = \sqrt{x} \longrightarrow F(2) = 1.4142$

$F'(x) = \frac{1}{2}x^{-\frac{1}{2}} \longrightarrow F'(2) = 0.3536$

$F''(x) = -\frac{1}{4}x^{-\frac{3}{2}} \longrightarrow F''(2) = -0.0884$

$F'''(x) = \frac{3}{8}x^{-\frac{5}{2}} \longrightarrow F'''(2) = 0.0663$

true value $\sqrt{2.5} = 1.5811$

2 terms → $F(x) = F(x_i) + \frac{F'(x_i)(x-x_i)}{1!}$

$= 1.4142 + 0.3536(2.5-2) = 1.5910$

$\epsilon_t = \left|\frac{1.5811 - 1.5910}{1.5811}\right| \times 100 = 0.62\%$

3 terms → $F(x) = F(x_i) + F'(x_i)(x-x_i) + F''(x_i)\frac{(x_i-x_i)^2}{2!}$

$= 1.5910 + \frac{-0.0884}{2}(0.5)^2 = 1.5800$

$\epsilon_t = \left|\frac{1.5811 - 1.5800}{1.5811}\right| \times 100 = 0.069\%$

4 terms → $F(x) = \ldots + F'''(x_i)\frac{(x-x_i)^3}{3!}$

$= 1.5800 + \frac{0.0663}{3\times2\times1} \times (0.5)^3 = 1.5813$

$\epsilon_t = 0.0126\%$ ∵ $\epsilon_t < 0.05$ ✓

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Q2)

False Position $x_m = x_u - \frac{f(x_u)(x_L - x_U)}{f(x_L) - f(x_U)}$	$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ **Newton-Raphson**

a) Consider $f(x) = e^{-x}(3.2\sin(x) - 0.5\cos(x))$

Use the **false position method** to solve the equation on the interval [3, 4] (3 iterations). Calculate the relative approximate error at each iteration. **(18 marks)**

#	xl	f(xl)	Xu	f(xu)	Xm	f(Xm)	Relative approximation error (RAE)
1	**3**	**0.04793**	**4**	**-0.03837**	**3.548**	**-0.02273**	**-**
2	**3**	**0.04713**	**3.548**	**-0.02273**	**3.369**	**-0.008066**	**5.313 %**
3	**3**	**0.04713**	**3.369**	**-0.008066**	**3.315**	**-0.002116**	**1.628 %**

b) For $f(x) = x^3 - 9x + 2$Use the Newton-Raphson method to solve the equation (3 iterations, $x_0 = 0$). Calculate the relative approximate error at each iteration.

(18 marks)

i	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$\lvert x_{i+1} - x_i \rvert$	RAE
1	**0**	**2**	**0**	**0.2222**	**0.2222**	**100%**
2	**0.2222**	**0.01117**	**-8.851**	**0.2234**	**0.0012**	**0.5371 %**
3	**0.2234**	**0.0005493**	**-8.85**	**0.2235**	**0.0001**	**0.045 %**

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$F(x) = x^3 - 9x + 2$

$F'(x) = 3x^2 - 9$

$x_{new} = x_i - \frac{F(x_i)}{F'(x_i)}$

itr. #1 $\Rightarrow$ $x_1 = 0 - \frac{F(0)}{F'(0)} = 0 - \frac{2}{(-9)} = 0.2222$ $\epsilon_a = 100\ \%$

#2 $\Rightarrow$ $x_2 = 0.2222 - \frac{F(0.2222)}{F'(0.2222)} = 0.2222 - \frac{(0.0110)}{(-8.8519)} = 0.2234$

$\epsilon_a = \left|\frac{0.2234 - 0.2222}{0.2234}\right| \% = 0.537\%$

#3 $\Rightarrow$ $x_3 = 0.2234 - \frac{F(0.2234)}{F'(0.2234)} = 0.2234 - \frac{(5.49 \times 10^{-4})}{(-8.8503)} = 0.2235$

$\epsilon_a = \left|\frac{0.2235 - 0.2234}{0.2235}\right| \% = 0.0447\%$

Q3) Solve the following linear system using LU decomposition **(25 marks)**

$x_1+x_2+x_3=1,$
$4x_1+3x_2-x_3=6,$
$3x_1+5x_2+3x_3=4$

$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$	$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & -10 \end{bmatrix}$	$Z = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$	$X = \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix}$

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Q4) a) Use the linearized data model to find the constants of the regression model $y = axe^{-bx}$ to best fit the following data set **(15 marks)**

x	1	2	3	4
y	2	3	4	4.5

b) Determine whether the curve a good fit for the data or not **(10 marks)**

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n}$$

$$S_t = \sum (y_i - \overline{y_{model}})^2$$

$$S_r = \sum_{i=1}^{n} (y_i - y_{model})^2$$

$$r = \sqrt{\frac{S_t - S_r}{S_t}}$$

a-

$y = axe^{-bx}$

$\frac{y}{x} = ae^{-bx} \Rightarrow \ln(\frac{y}{x}) = \ln a - bx \ln e$ $\quad \ln e = 1$

$\Rightarrow \underbrace{\ln(\frac{y}{x})}_{y_n} = \underbrace{\ln a}_{a_0} \underbrace{- b}_{a_1} x$

$\Rightarrow$ model $y_n = a_0 + a_1 x$

x	x_1	1	2	3	4
$y_n = \ln(\frac{y}{x})$		0.6931	0.4055	0.2877	0.1178

$\Sigma x = 10$ $\quad \Sigma y = 1.5041$

$\Sigma x^2 = 30$ $\quad \Sigma xy = 2.8383$

$a_1 = \frac{n\Sigma xy - \Sigma x \Sigma y}{\Sigma x^2 - (\Sigma x)^2} = -0.1844 \longrightarrow$ (b) $a_{1m} = 0.1844$

$a_0 = \frac{\Sigma y}{n} - a\frac{\Sigma x}{n} = 0.8370 \longrightarrow$ (a) $a_{0m} = e^{a_0} = 2.3094$

the model: $y_m = 2.3094 \times e^{-0.1844x}$

b- Sr= 0.05118, St= 3.688 → r = 0.993 → the curve a good fit for the data